

Keyword News Aggregator

A Non-Technical Overview of Current Architecture, Optimization Findings, and Performance Strategy

■ ■ The Good News: Strong Engine • Core systems are rock solid: RSS feeds, Supabase database, and Mistral AI summaries run with high reliability. • Sleek modern presentation: Dark mode, layout animations, and responsive cards create a high-end reader experience. • Fully automated SEO: Search-engine pre-rendering and dynamic sitemaps are active on Netlify.		■ ■ The Opportunity: Spring Cleaning • Unused packages: The website is packing and downloading tools it never actually uses (dead weight in the trunk). • Double math: Every topic tap triggers a phantom second recalculation of the news feed. • Component sprawl: Three massive files are doing 5 jobs at once instead of having specialized helpers.
CURRENT RATING B+ Strong core, ready for clean-up	TARGET SAVINGS ~100 KB+ Prune ghost packages & lean types	FEED SPEED Instant (1-Pass) Eliminate double-render recalculations

Detailed Audit Findings (In Plain English)

Understanding the areas of friction in your current software and how fixing them benefits your users.

Finding 1: Dead Baggage in the Trunk UNNECESSARY DOWNLOAD PACKAGES	The website currently packs two software packages (rss-parser and uuid) into its download bundle that are never imported or used anywhere. Every mobile visitor downloads dozens of kilobytes of ghost code before the feed appears. The Fix: Delete the unused tools immediately with zero visual or functional risk.
Finding 2: Doing the Same Math Twice DOUBLE-RENDERING ON TOPIC CLICKS	Whenever a reader switches topics or searches, the app displays the articles and then immediately reruns the exact same clustering and ranking math a phantom second time in the background. The Fix: Convert the calculation to <i>'Calculate once and display instantly'</i> (via useMemo) to save battery and boost responsiveness.
Finding 3: Overcrowded Rooms MONOLITHIC COMPONENT FILES	Three components (Article Page, Market Pulse Ribbon, and Article Cards) are doing 5 jobs at once. For example, every single news card on screen loads its own hidden AI summary popup in memory, even if the reader never clicks it. The Fix: Split large files into focused helpers and load popups on demand.
Finding 4: Developer Sticky Notes BACKGROUND CHATTER & 35 WARNINGS	The automated code quality scanner flagged 35 errors/warnings (unused imports, empty error handlers). Additionally, 9 developer console.log statements print messages to the console on every click. The Fix: Silence production logs and clean all 35 warnings for a 100% clean, professional build.
Finding 5: The Cluttered Front Porch ROOT DIRECTORY DOCUMENTATION CLUTTER	Old Python book generator scripts and 170 KB of reference PDF handbooks sit in the main project folder next to website code, making git checkouts heavier than necessary. The Fix: Move all non-production manuals and generator scripts into a clean docs/handbook/archive.

Action Plan & Business Payoff

A structured, low-risk roadmap to achieve an ultra-lean, high-performance web app.

Area	Current State (Before)	Optimized State (After)
Download Weight	Carries 2 dead packages and bulky types	Lean download (~100 KB+ lighter)
Topic Switching	Recalculates feed twice per click	Instant, single-pass render
Memory Footprint	30 hidden modals in device RAM	Modals created only when tapped
Code Warnings	35 warnings and active console chatter	0 warnings, silent production console
Future Upgrades	Harder to modify 680-line mixed files	Clean, modular components

Step 1: The Quick Win (15 Mins) • Delete dead packages (rss-parser, uuid) • Clear all 35 code warnings & dead imports • Silence production console.log spam <i>Result: 100% clean, error-free build with zero visual changes.</i>	Step 2: Engine Tune-Up (30 Mins) • Convert feed filtering to single-pass useMemo • Stop double calculations on topic clicks • Memoize providers to stop wasted re-renders <i>Result: Snappier navigation, cooler device running, lower battery drain.</i>	Step 3: Room Makeover (Modular) • Split ArticlePage into Sentiment & Explainer helpers • Separate Market Ribbon from Watchlist Modal • Move reference PDFs & scripts to docs/archive <i>Result: Clean architecture that is joyful to maintain and extend.</i>
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